Pathway and network analysis

An introduction to pathway analysis with WikiPathways

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Computational Systems Biology for Digital Medicine, WCS course 2025
08 December 2025

Interactive session

- Introduction
- Explanation and exploration of the dataset
  - Differential gene expression
  - Small cell lung cancer
- Pathway enrichment analysis
  - Pathway databases
  - Statistical analysis to identify affected pathways in dataset
- Pathway visualization
  - Data visualization and interpretation

Setup

- This will be an interactive session with switching between lecture / demo and practical steps.

- Read the explanations of the steps and discuss the output
$\rightarrow$ try to understand and interpret the results

- **Discuss with each other**

Introduction

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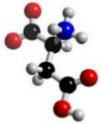
Learning objectives

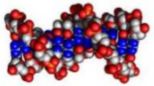
1. Study the molecular mechanisms underlying Small Cell Lung Cancer
2. Biological insights into lung cancer from transcriptomics data
3. Tools for pathway analysis and visualization

Introduction

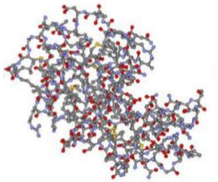
Molecules of life do not function in isolation ....



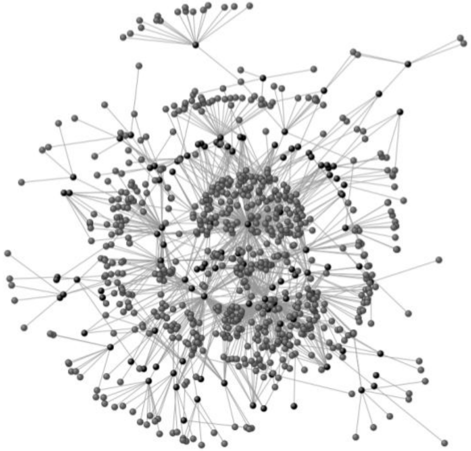
metabolites



genes

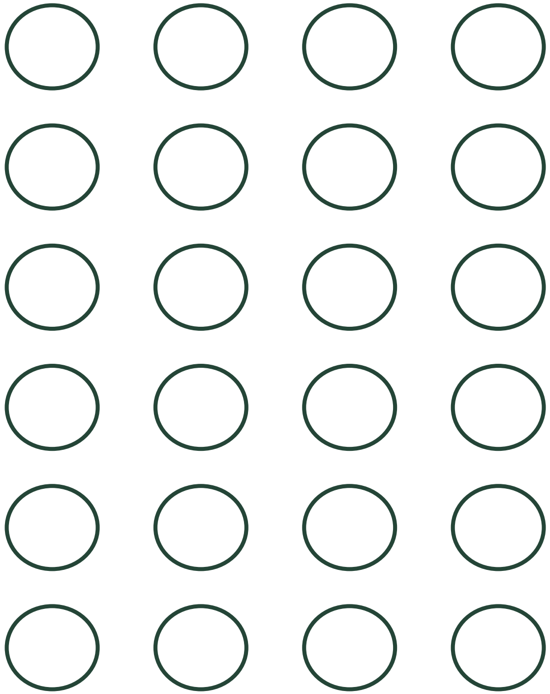


proteins



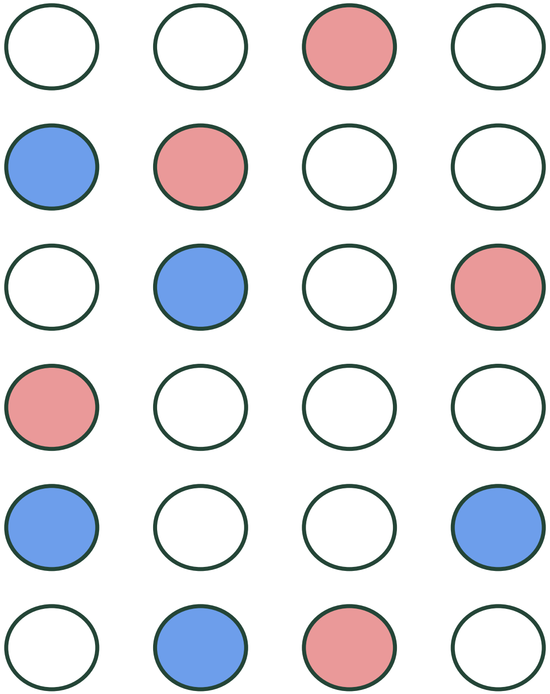
but form complex networks that define a cell ....

Introduction enrichment analysis



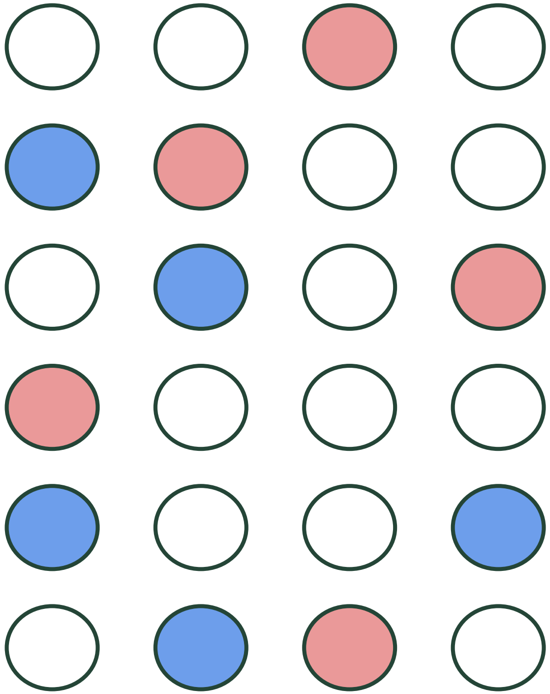
Quantify
Isolated data points

Introduction enrichment analysis



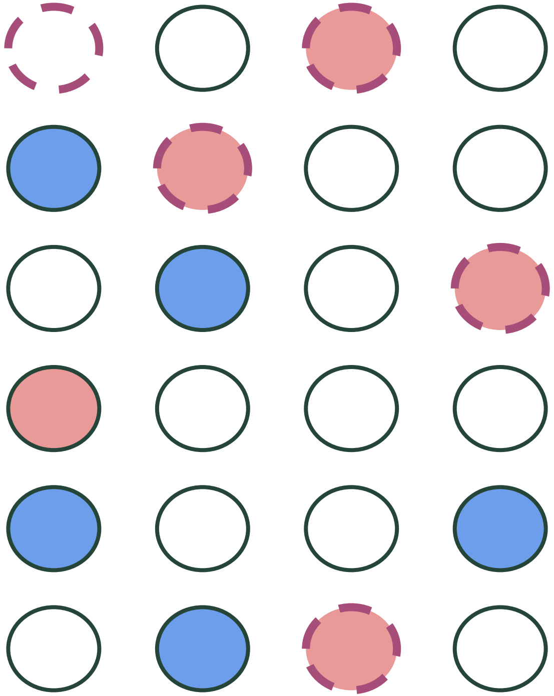
Comparative statistics
Genes of interest
(up- or down-regulated genes)

Introduction enrichment analysis



Enrichment analysis
Pre-defined gene sets → functional groups

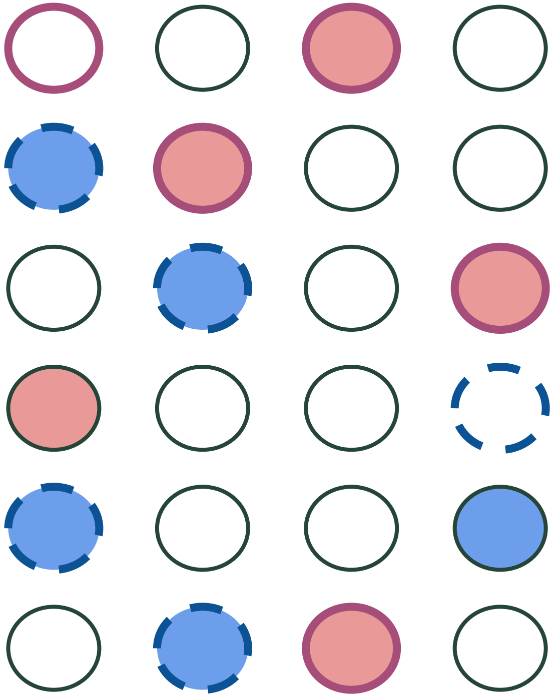
Introduction enrichment analysis



Enrichment analysis
Pre-defined gene sets → functional groups

Apoptosis

Introduction enrichment analysis

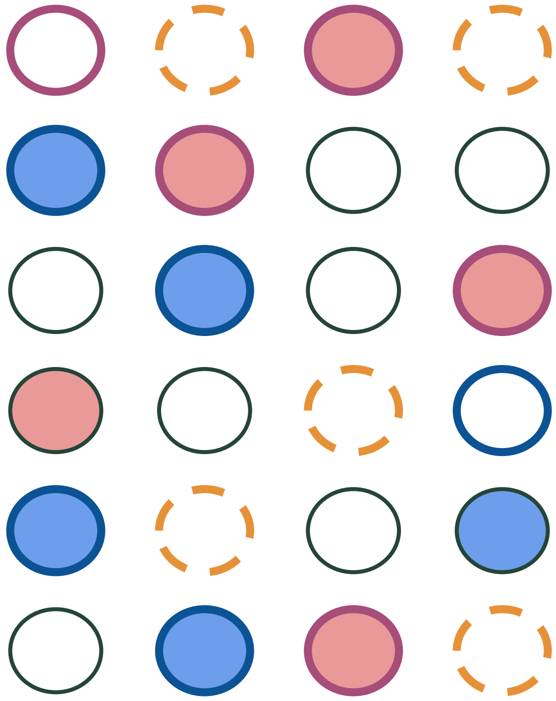


Enrichment analysis
Pre-defined gene sets → functional groups

Apoptosis

Amino acid metabolism

Introduction enrichment analysis



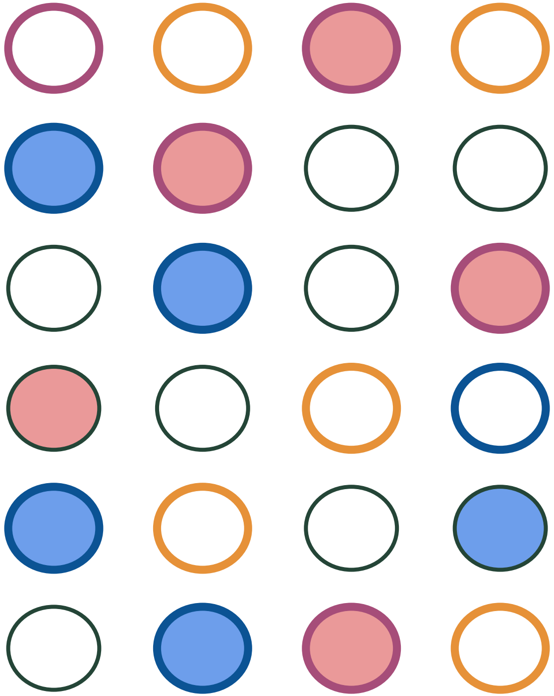
Enrichment analysis
Pre-defined gene sets → functional groups

Apoptosis

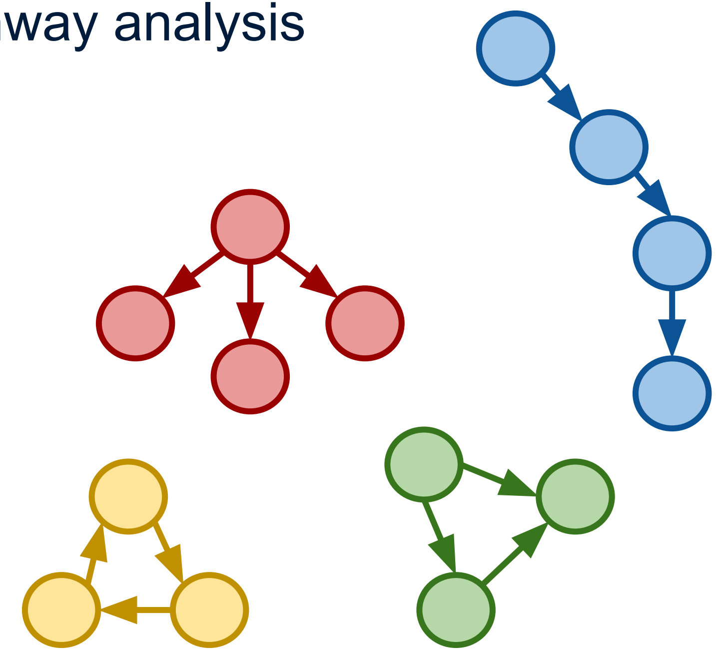
Amino acid metabolism

IL6 signaling pathway

Introduction enrichment analysis



Enrichment analysis
Pathway analysis



Pathway = gene set with information about relationships

Explanation and exploration of the dataset

Dataset

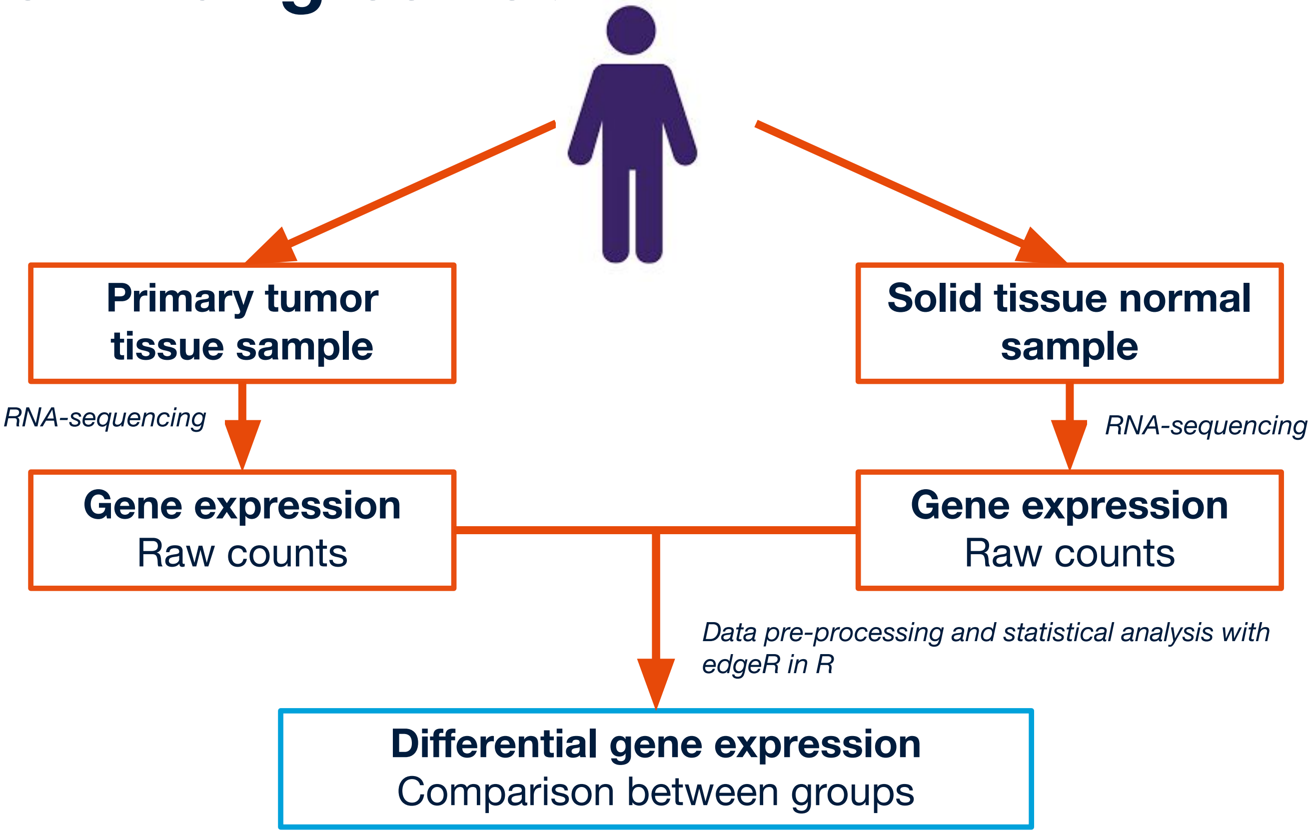
- Jiang, *et al.* (2016) The RNAseq of 79 small cell lung cancer (sclc) and 7 normal control

- Publicly available dataset - GEO database
  - https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE60052

- Comparing SCLC vs. healthy control

Jiang L, Huang J, Higgs BW, Hu Z, Xiao Z, Yao X, Conley S, Zhong H, Liu Z, Brohawn P, Shen D.
Genomic landscape survey identifies SRSF1 as a key oncodriver in small cell lung cancer.
*PLoS genetics.*2016 Apr 19;12(4):e1005895.

Dataset – lung cancer



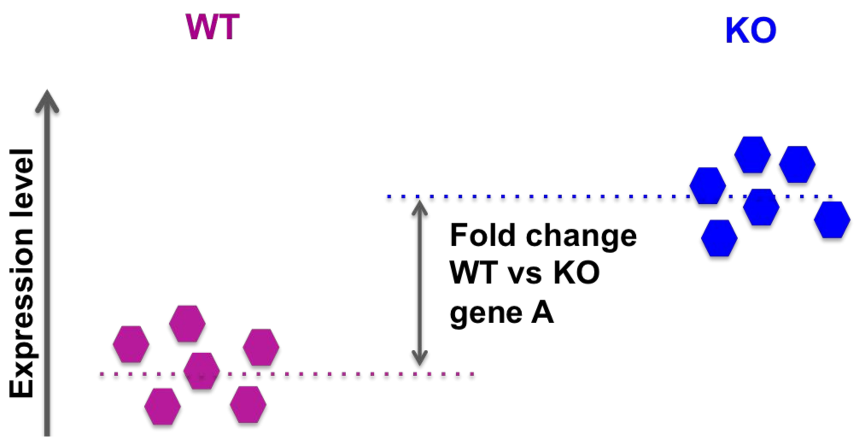
Dataset

GeneID	GeneName	log2FC	pvalue	padj
ENSG00000230657	PRB4	13.2739759	0.003920605	0.097861917

- GeneID □ identifier in online database
- GeneName □ official gene symbol
- log2FC □ log2 of fold change (ratio of the differences between cancer and healthy samples)
- pvalue □ significance level of comparison
- padj □ corrected P.Value for multiple testing

Dataset

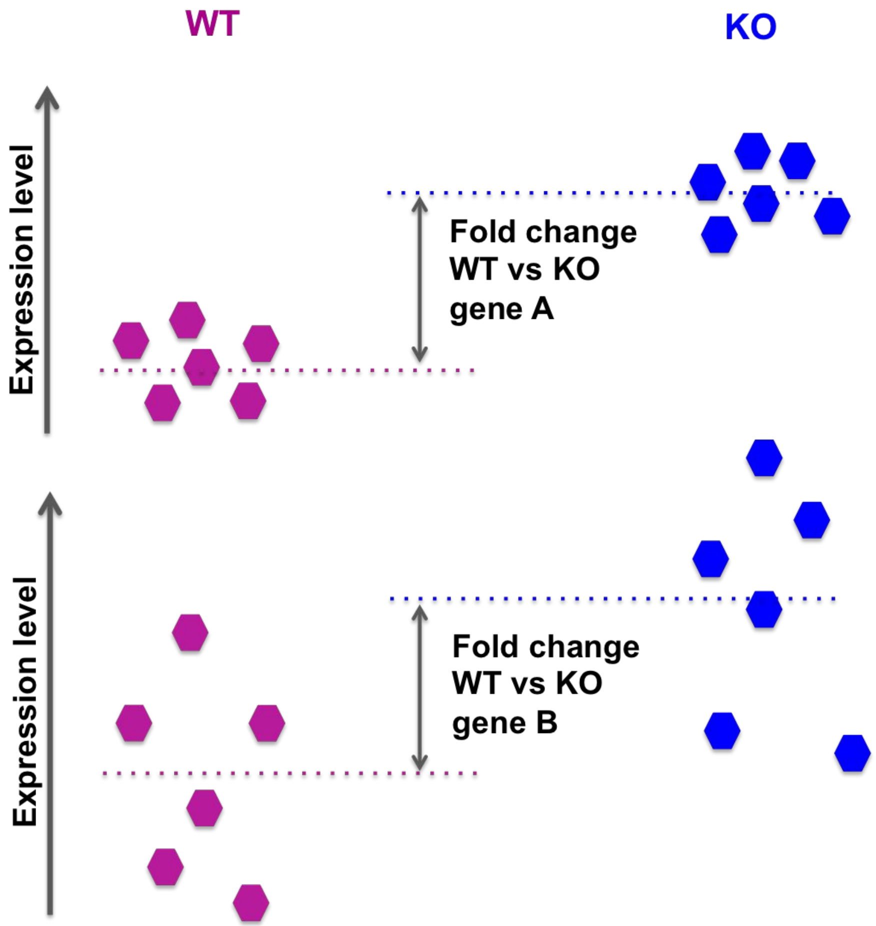
What is the fold change?



Source: https://biocorecrg.github.io/CRG_Bioinformatics_for_Biologists/differential_gene_expression.html

Dataset

What is the fold change?



Source: https://biocorecrg.github.io/CRG_Bioinformatics_for_Biologists/differential_gene_expression.html

Dataset - log2 fold change

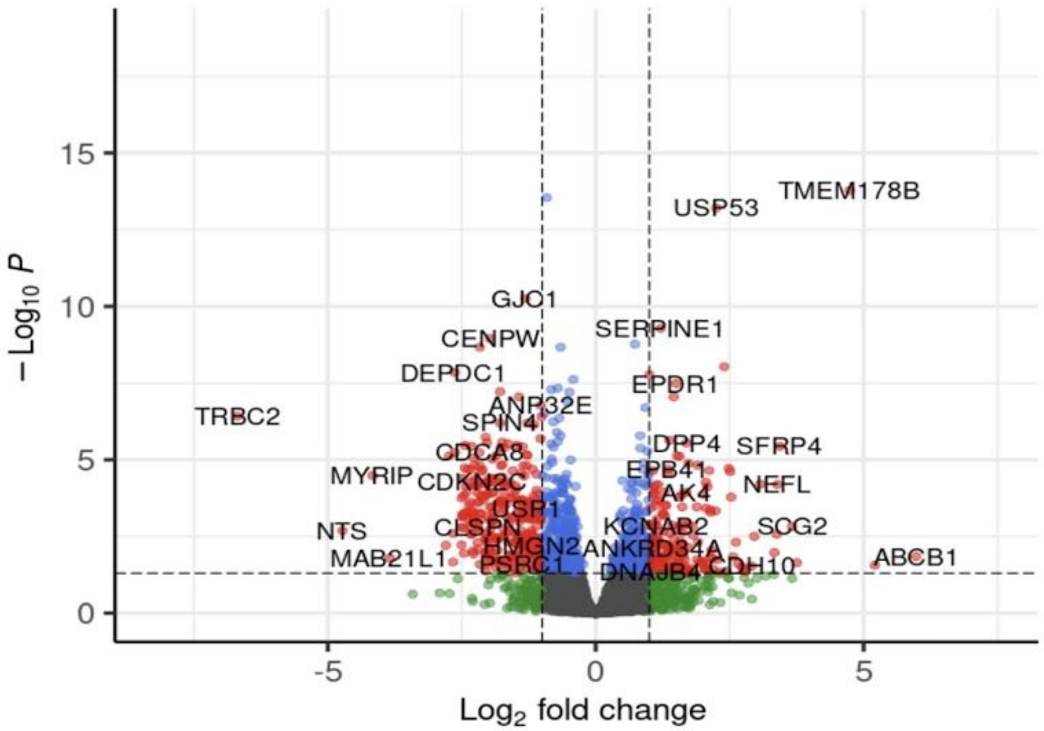
- Is the gene more or less expressed in cell senescence compared to control?

Negative log2FC (< 0) downregulation in cell senescence	**log2FC = 0** both group show same expression level	**Positive log2FC (>0)** upregulation in cell senescence

Example: log2FC = 1 → fold change = 2^1 = 2 = 2-fold increase

- gene is twice as expressed in cell senescence compared to control

Dataset - Volcano plot



Typical cutoff values:
log2FC = 1.00
log2FC = 0.58
log2FC = 0.26

Hands-on Module 1 - Explore the dataset

Exploration:

- lung cancer vs. healthy

Questions:

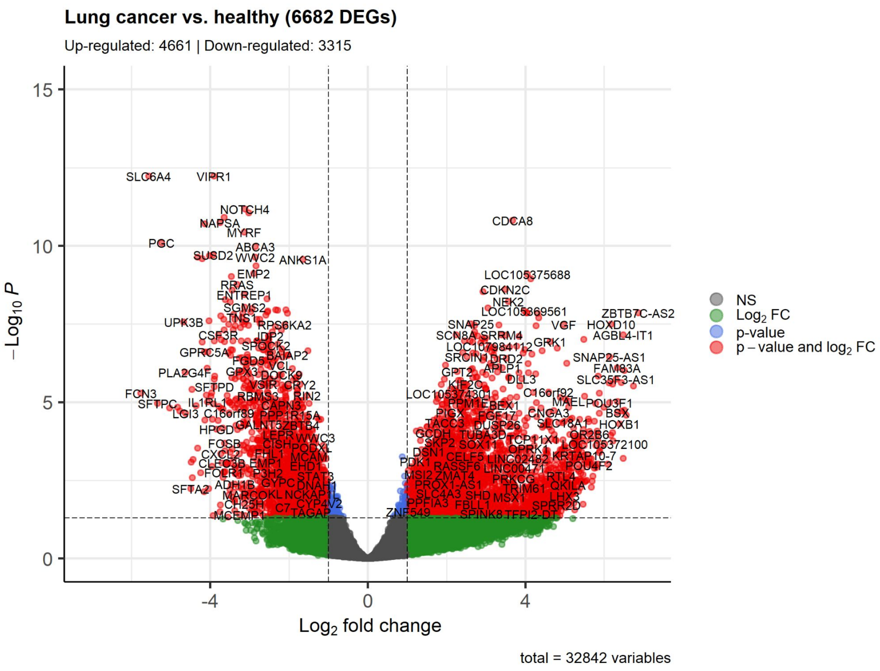
- Number of differentially expressed genes ($|\log2FC| > 1$ and $p\text{-val} < 0.05$)?
- Number of up-regulated genes?
- Number of down-regulated genes?
- Create the volcano plot → interpretation (discuss with each other)

Key insights

- More up-regulated than down-regulated genes

- We're looking at ~32,000 genes in total and nearly 7,000 genes that are differentially expressed - that's too many to interpret individually

- Volcano plot shows both magnitude (x-axis) and confidence (y-axis) of difference!



Pathway statistics
Enrichment analysis

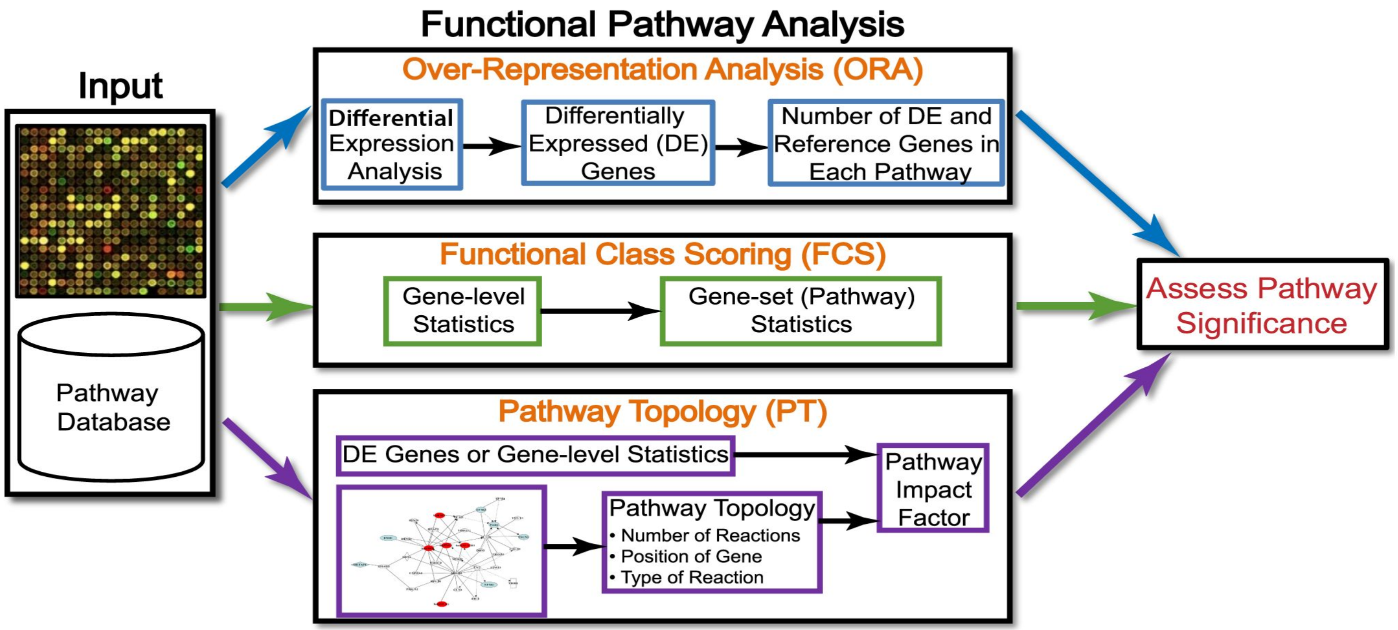
Enrichment analysis

- Aims:
  - Put data into a biological context
  - Intuitive, simple and reproducible analysis
  - Provide a visual representation of the results
  - Reduce complexity from large datasets (thousands of genes) to a set of affected processes

What pathway do we expect to be affected?

- Loss of normal growth control leads to widespread changes
  - Uncontrolled proliferation and survival
  - Resistance to cell death signals
  - Altered metabolism and energy production
  - Tissue invasion and metastatic spread
  - Recruitment of blood vessels (angiogenesis)
  - Evasion of immune surveillance
  - …

Pathway enrichment analysis



Over-representation analysis

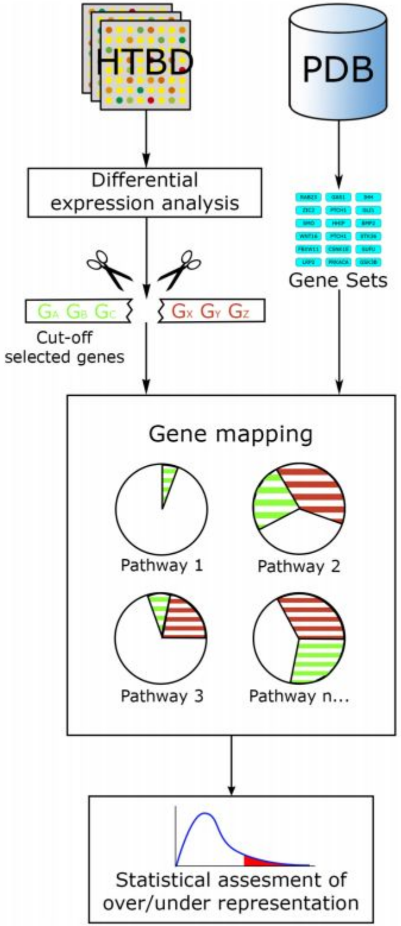
- **Input**
  1. Transcriptomics dataset
     - Background = all measured genes
     - Input list = significantly changes genes
  2. Pathway database

WikiPathways demo

Over-representation analysis

- **Input**
  1. Transcriptomics dataset
     - Background = all measured genes
     - Input list = significantly changes genes
  2. Pathway database

- **Statistical test** (hypergeometric test)



Source: García-Campos (2015) doi: 10.3389/fphys.2015.00383

Over-representation analysis

- **Input**
  1. Transcriptomics dataset
     - Background = all measured genes
     - Input list = significantly changes genes
  2. Pathway database

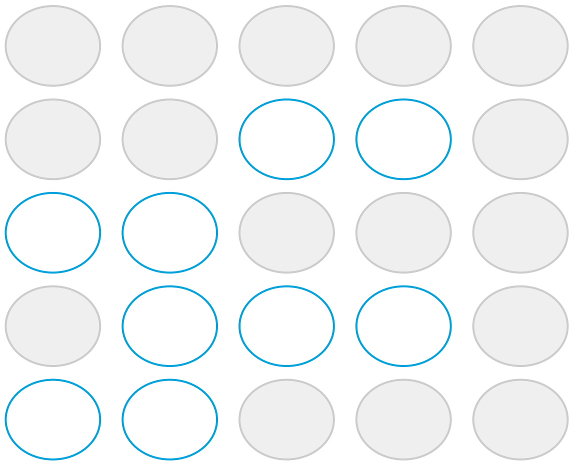
- **Statistical test** (hypergeometric test)

- **Output**
  - Enrichment score / p-value per pathway → ranked pathway list

Over-representation analysis

- **N = 25**
  background list (total number of measured genes in experiment)

Over-representation analysis



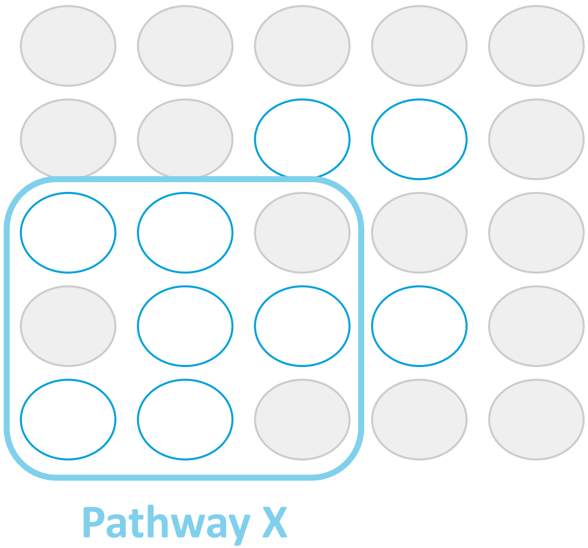
- **N = 25**

  background list (total number of measured genes in experiment)

- **R = 9**

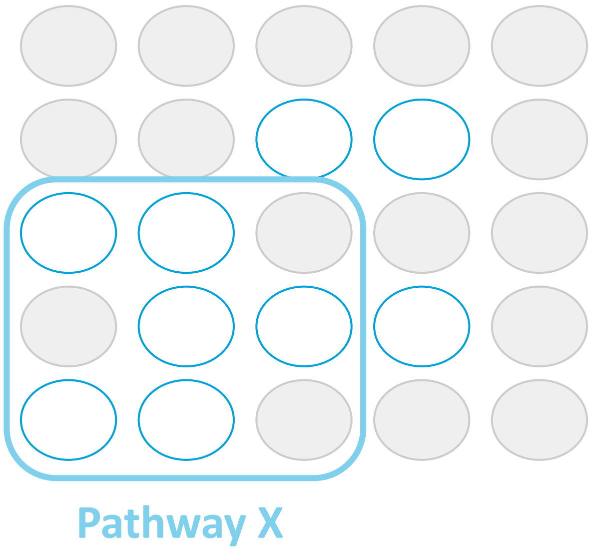
  input list (number of changed genes in experiment)

Over-representation analysis



- **N = 25**
  background list (total number of measured genes in experiment)
- **R = 9**
  input list (number of changed genes in experiment)
- **n = 9**
  total number of genes in pathway

Over-representation analysis



- **N = 25**
  background list (total number of measured genes in experiment)
- **R = 9**
  input list (number of changed genes in experiment)
- **n = 9**
  total number of genes in pathway
- **r = 6**
  number of changed genes in pathway

Pathway analysis results

- Overrepresentation analysis:
  - Ranked list of pathways

Table 1 Seven pathways changed in the diabetic liver

Pathway	Z-score	P-value	# Genes	Genes
Triacylglyceride Synthesis	3.78	0.001	3 / 19	↑ AGPAT2, GPD1, DGAT1
Proteasome Degradation	3.32	0.006	5 / 53	↑RPN1, PSMB3, HLA-B, HLA-E, HLA-J
Statin Pathway	3.10	0.006	3 / 25	↑DGAT1, APOA4, CYP7A1
Fluoropyrimidine Activity	2.84	0.013	3 / 28	↑SLC22A7 ↓ABCG2, DPYD
Pathogenic Escherichia coli infection	2.76	0.011	4 / 46	↑ ARPC1A, ARPC1B, ACTB ↓ ROCK1
Adipogenesis	2.41	0.016	7 / 121	↑ SREBF1, CDKN1A, NR1H3, PNPLA3, AGPAT2 ↓ CISD1, ZMPSTE24
AMPK Signaling	2.38	0.029	4 / 54	↑ SREBF1, P21 ↓ LEPR, PFKFB3

Pathway statistics in PathVisio revealed seven significantly altered pathways (Z-score > 1.96, P-value < 0.05, minimum of 3 changed genes). The number of genes (# Genes) represent the number of differentially expressed genes in the pathway compared to the total number of measured genes in the pathway. The arrows indicate up ↑ and down-regulation ↓.

Source: https://doi.org/10.1186/1471-2164-15-971

Interpretation!

Be aware!!

- Be aware!
  - ORA **does not** take pathway topology into account!
  - You don’t know yet where the changes occur in the pathway.
  - Always look at the pathway diagrams and study the changes to make the right conclusions!

Hands-on Module 2 - Enrichment Analysis

- Perform pathway statistics to identify which pathways in WikiPathways are significantly affected

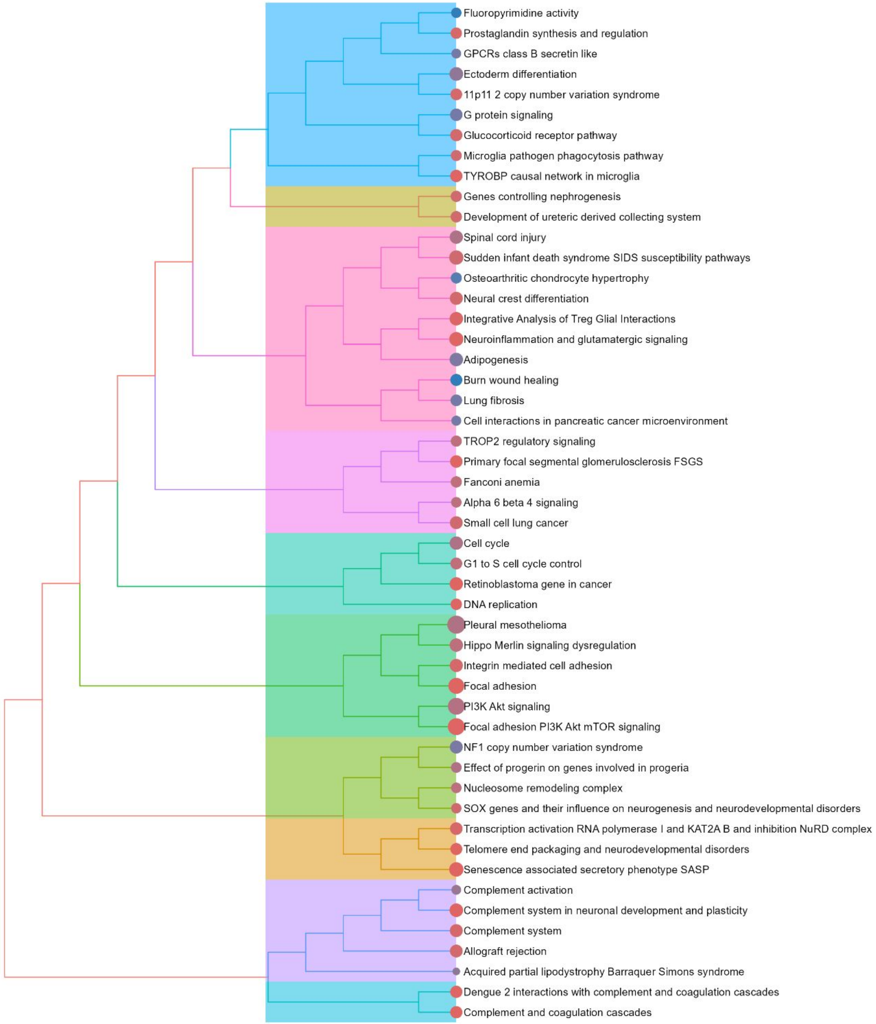
- **Questions**:
  - What pathways are affected?
  - Do you see the expected biological pathways dysregulated?

Key insights

- Enrichment asks: do we see more DEGs in this pathway than we'd expect by chance?

- Multiple interconnected pathways characterize SCLC biology

- **Proliferation**: Cell cycle control, G1 to S transition, DNA replication
- **Cell survival & migration**: PI3K-Akt signaling, focal adhesion, integrin-mediated adhesion
- **Neuroendocrine identity**: Neural crest differentiation, neurogenesis
- **Tumor microenvironment**: SASP, complement activation, immune interactions
- **Development**: SOX genes, telomere maintenance, nucleosome remodeling
- …



Pathway visualization

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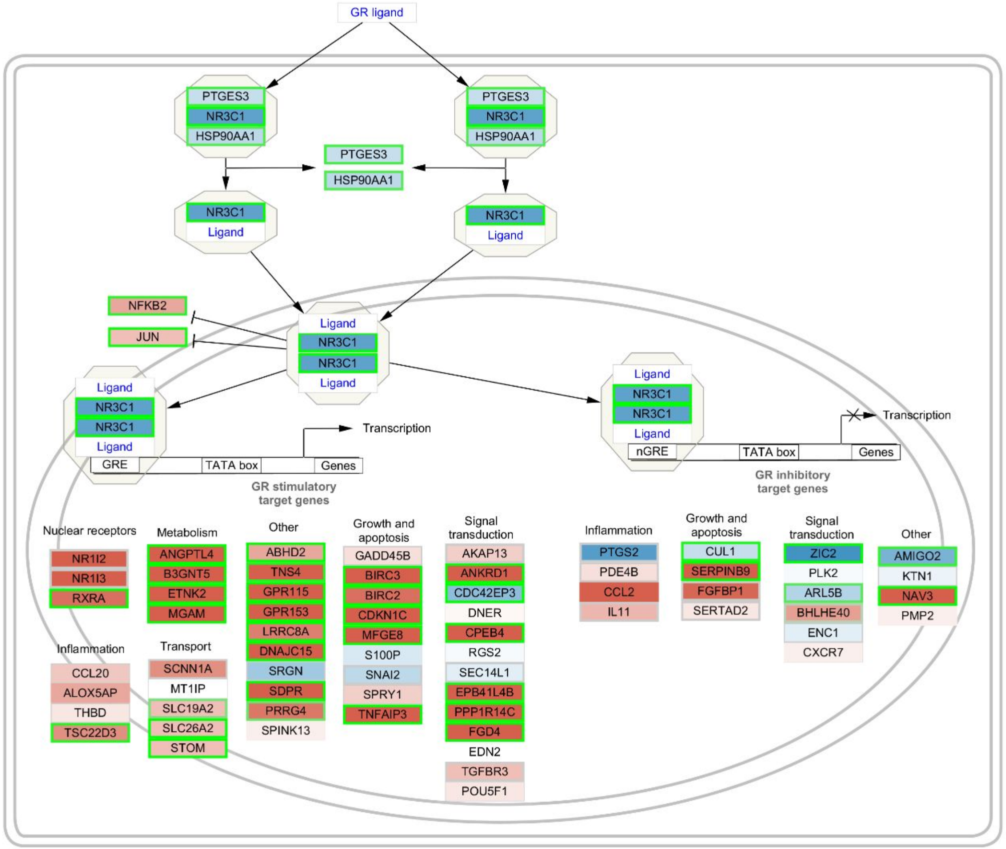
Demo Cytoscape

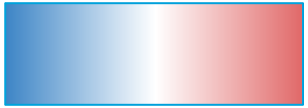
1. Open pathway in Cytoscape (example: WP5321)

2. Import data

3. Style pathway

Pathway visualization





log2FC

Hands-on Module 3 - Pathway visualization

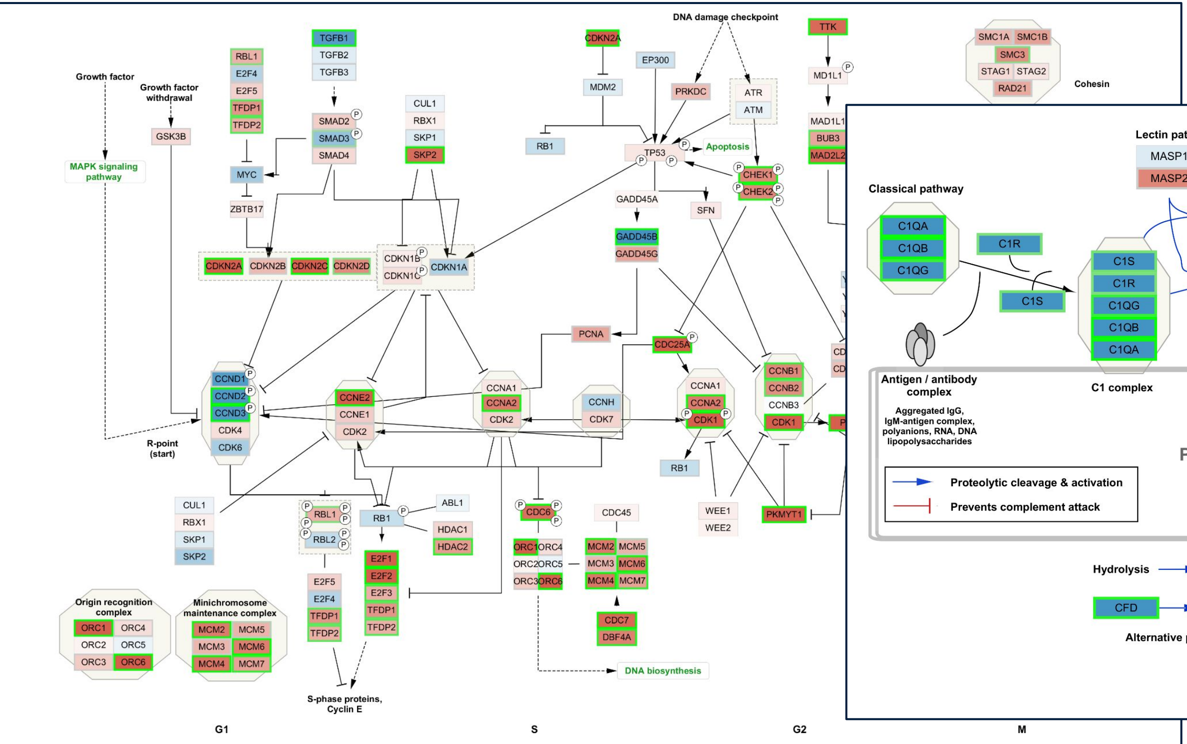
- Make sure you have Cytoscape open

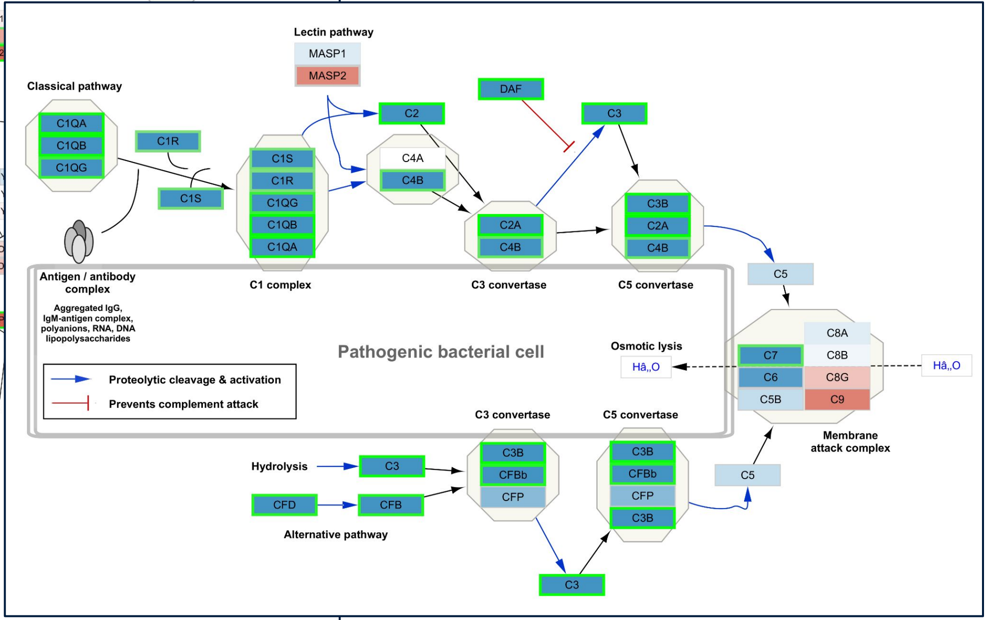
- Script will open pathways in Cytoscape, add data and create a visual style

- **Questions**:
    - Which genes in the pathway are up- or down-regulated?
    - Does this make sense biologically?

Key insights

- Cell cycle strongly up-regulated
- Complement activation strongly down-regulated





Interactive session

- Introduction
- Explanation and exploration of the dataset
  - Differential gene expression
  - Small cell lung cancer
- Pathway enrichment analysis
  - Pathway databases
  - Statistical analysis to identify affected pathways in dataset
- Pathway visualization
  - Data visualization and interpretation

What can you do with pathway models?

1. Data visualization
    - How is the pathway affected in my experiment?

2. Pathway statistics / enrichment analysis
    - Which pathways are affected in my experiment?

3. Network analysis (optional extension)
    - How is the pathway regulated by transcription factors / drugs?
    - How is the pathway connected to other pathways (pathway crosstalk)?

Summary

- Pathways are **intuitive, machine-readable models** of biological processes
- Pathways can be used to **analyze experimental data** (visualization and statistics)
- **Automation** (e.g. in R) makes an analysis reproducible and can be reused for another dataset
- There are strong changes in cancer cells on gene expression level!

Questions?

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